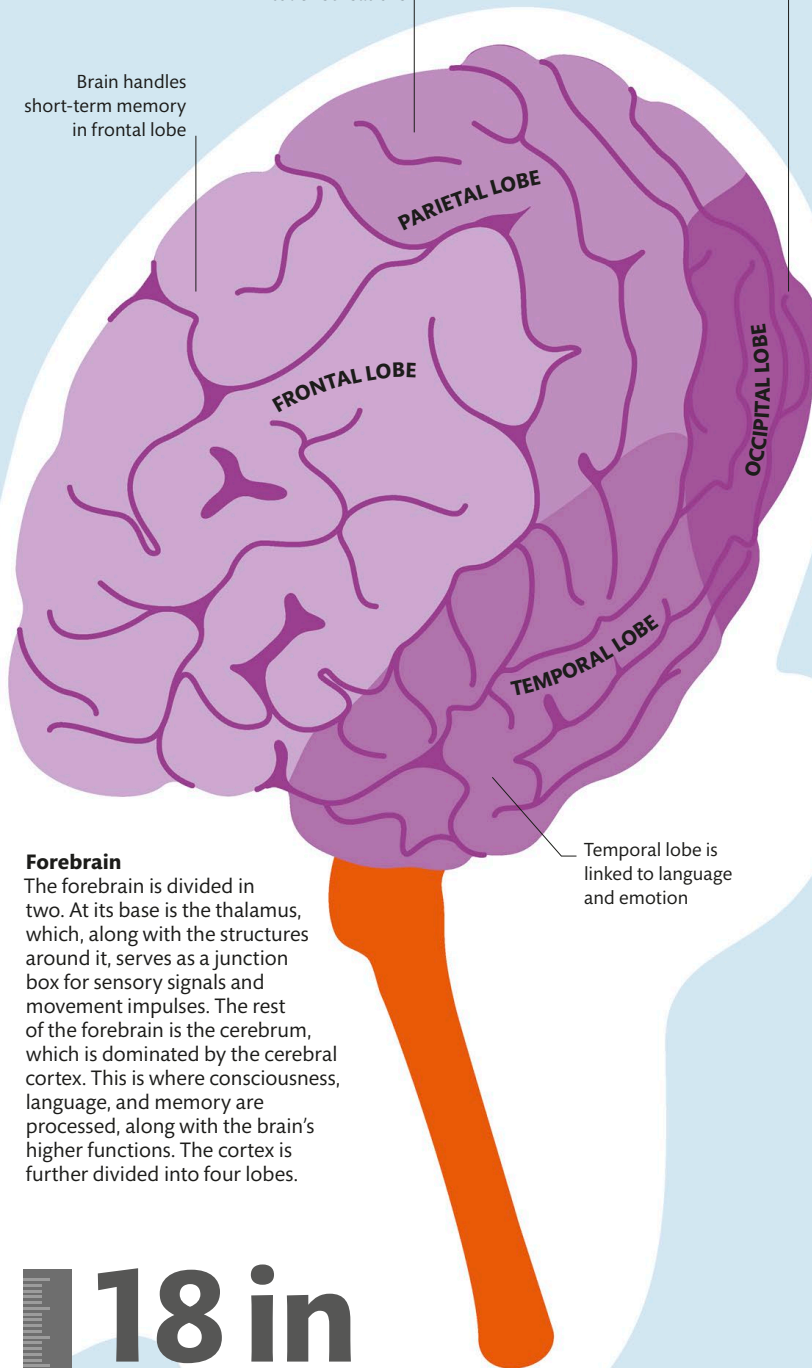




Parietal lobe governs perception of body position and other touch sensations

Occipital lobe is mostly given over to vision

Brain handles short-term memory in frontal lobe



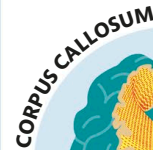
Forebrain

The forebrain is divided in two. At its base is the thalamus, which, along with the structures around it, serves as a junction box for sensory signals and movement impulses. The rest of the forebrain is the cerebrum, which is dominated by the cerebral cortex. This is where consciousness, language, and memory are processed, along with the brain's higher functions. The cortex is further divided into four lobes.

Temporal lobe is linked to language and emotion

Hemispheres

The cerebrum forms in two halves, or hemispheres, which are divided laterally by a gap called the longitudinal fissure. Nevertheless, the hemispheres share an extensive connection via the corpus callosum. Each hemisphere is a mirror image of the other, although not all functions are performed by both sides (see p.10). For example, speech centers tend to be on the left side.

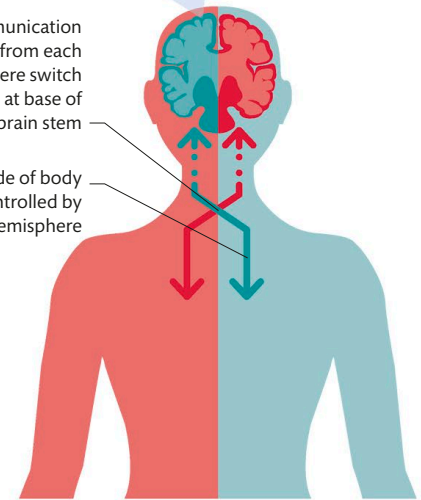


White-matter nerve tracts form corpus callosum

Same layout of four lobes on both sides

Communication fibers from each hemisphere switch sides at base of brain stem

Left side of body is controlled by right hemisphere



Left and right

The brain and the body are connected contralaterally, meaning that the left brain hemisphere handles the sensations and movements of the right side of the body and vice versa.

18 in
(46 CM) THE LENGTH
OF THE SPINAL CORD